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The Augmented Reality Protein Dynamics

Problem Statement

- Textbook depictions of 3D protein structures are **difficult** for biology students at WTCC to understand and visualize.
- This **AR application** will allow students to **visualize protein molecules** virtually in a **3D and interactive** way.

Functional Requirements

User roles

Admin

- Users

Student

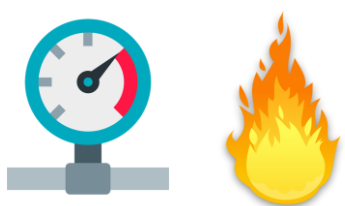
- AR View
- Lesson Plan Session

Instructor

- Molecules
- Environmental Factors + Tags
- AR View
- Lesson Plans

Store protein molecules

- uploaded using **.cif files**
- define the **environmental factors**



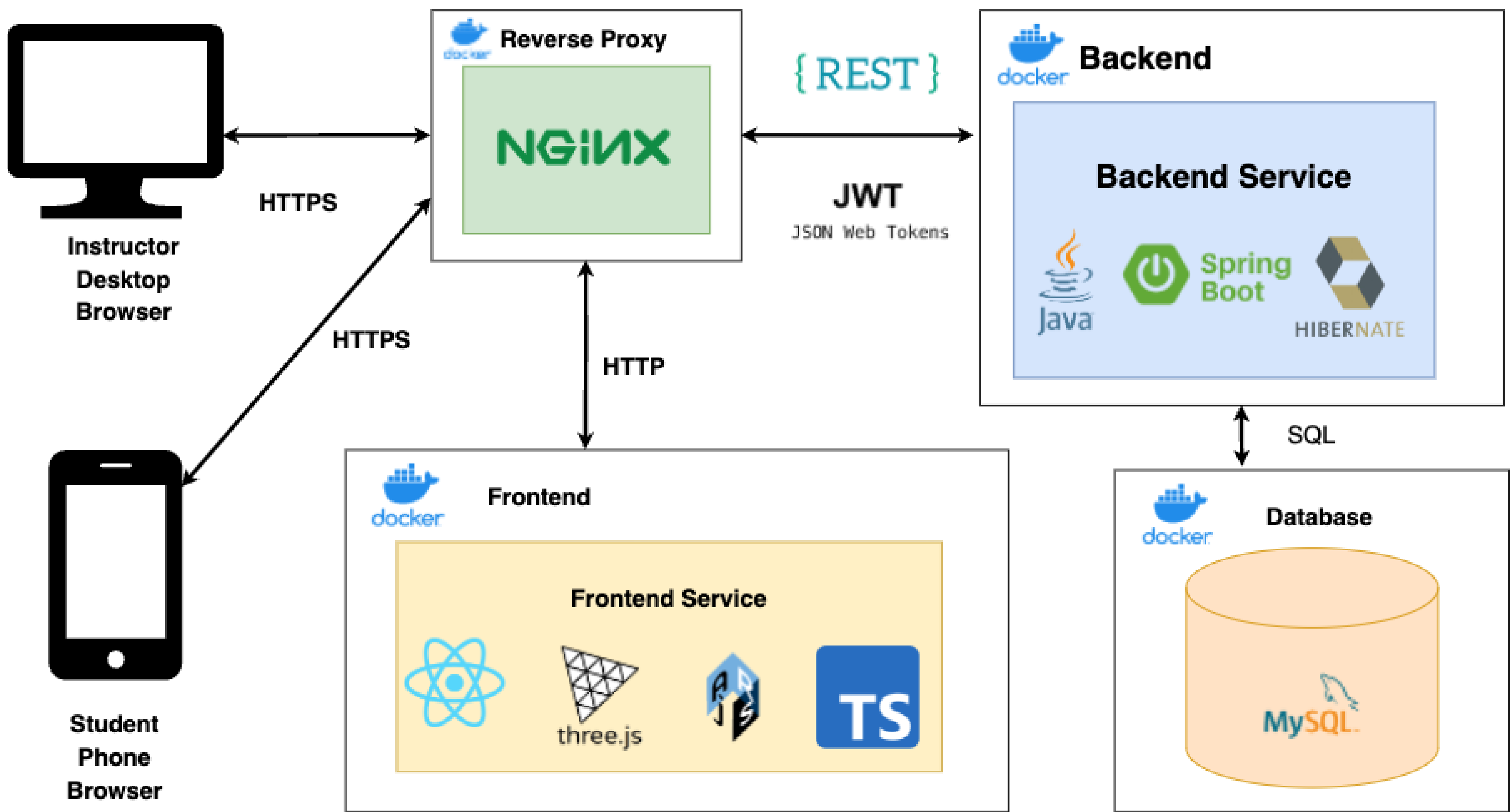
AR View

- Display a **3D structure** of the **protein molecule** when shown a valid **AR tag**.
- Apply **environmental factors** through additional **AR tags**.

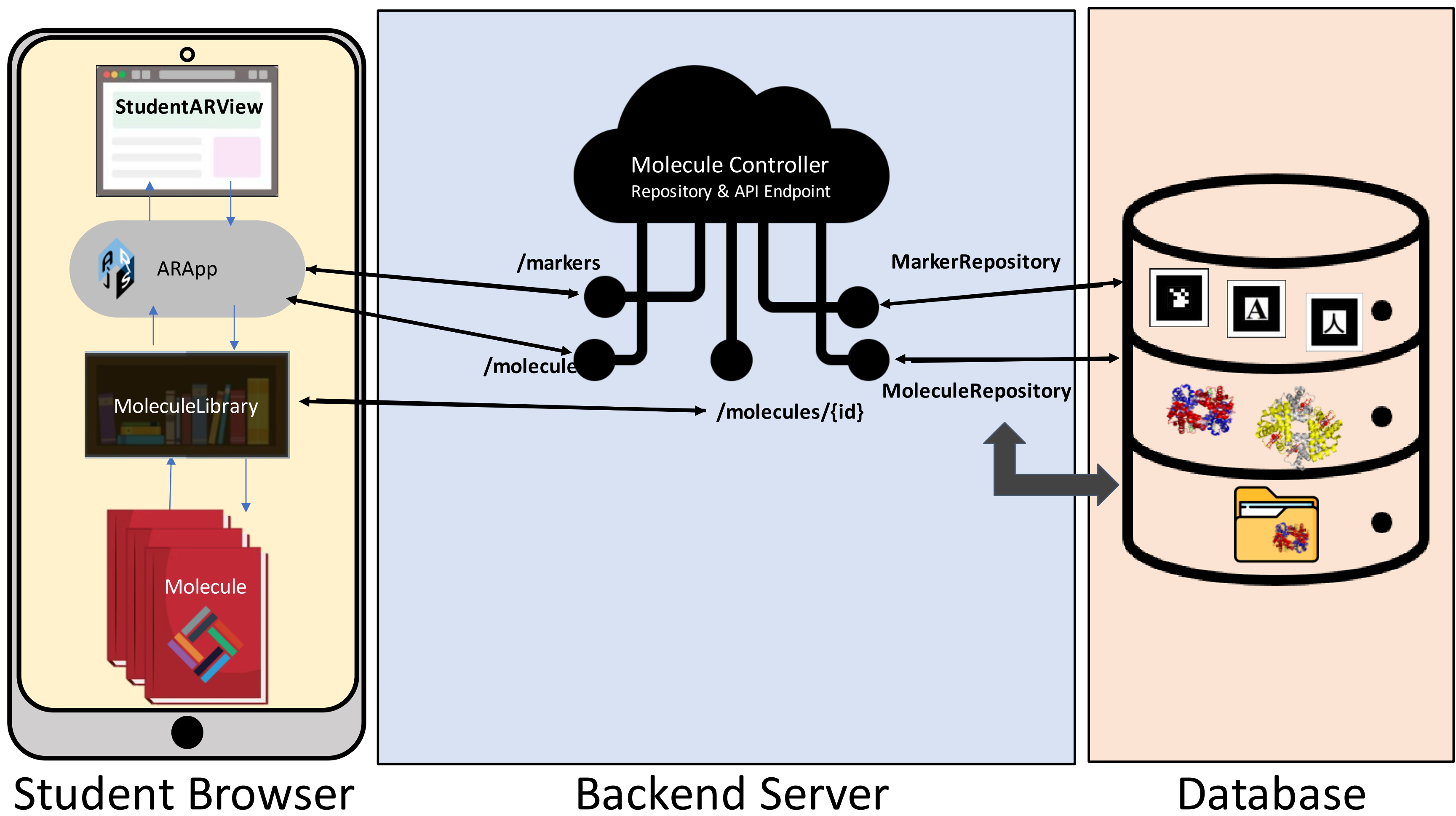
Lesson Plans

- Prompt **questions** when investigating the 3D structure
 - multiple choice
 - short answers
- Join **sessions** and submit **responses** to questions

High Level Design

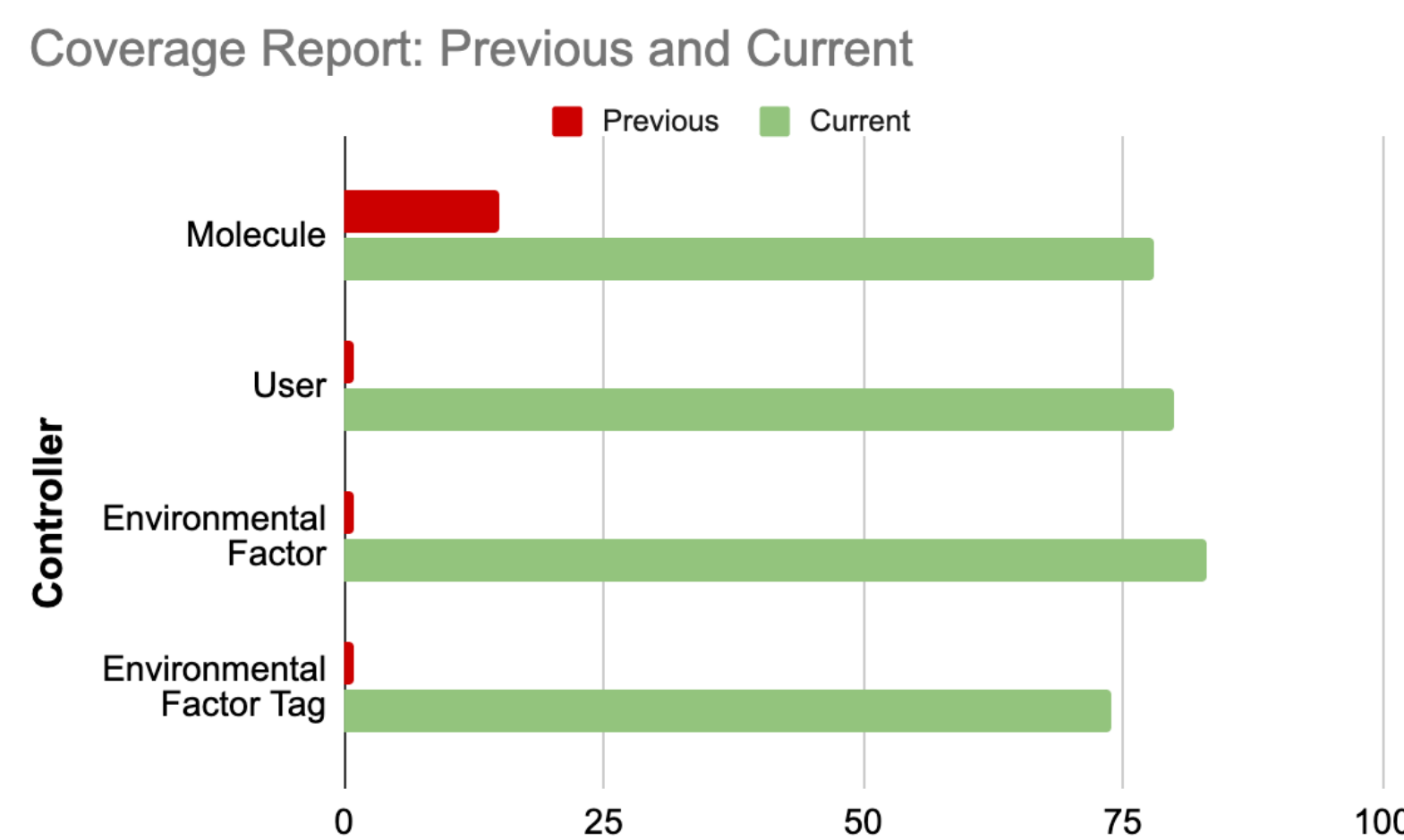


Low Level Design



Testing *

Testing Coverage Report (%)



Black Box Testing Updates

Molecules	Previous: Create Current: Create, Delete, Update, Read
Users	Previous: None Current: Create, Delete, Read
Environmental Factors	Previous: Delete, Update Current: Create, Delete, Update, Read
Lesson Plans	Previous: Create, Delete, Update, Read Current: No Changes

Black Box Tests Passing

Controller	# Passing / Total
Molecule	11 / 11
EF	4 / 4
User	3 / 3
Other	20 / 35

* Ask us about our updated tests

